STRUCTURE OF PROKARYOTIC CELLS

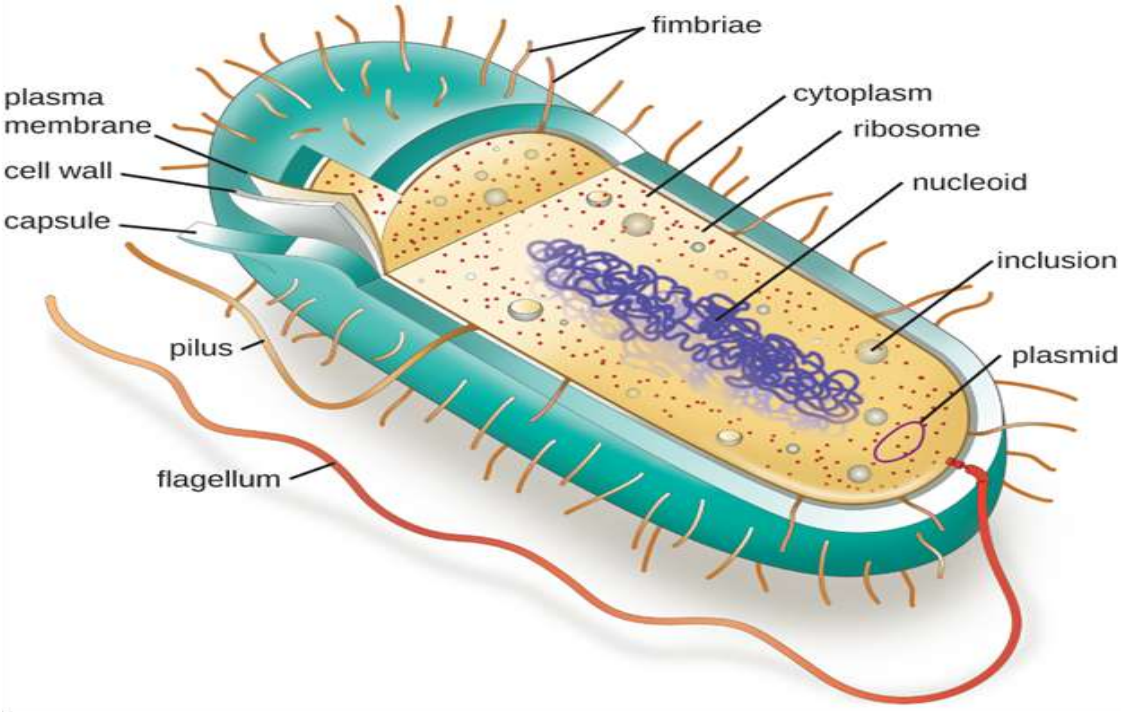
Prokaryotes

- All the living organisms in this world can be broadly divided into two groups based on their cell nucleus
- All those organisms that have a **well-defined nucleus** with a nuclear membrane enclosing the genetic material are called the **eukaryotes** and those which do not have a nuclear membrane covering their **genetic material** (DNA) are called the **prokaryotes**
- The prokaryotes include all the bacteria.
- Bacteria lack a nucleus and their genetic material freely floats in the cell
- These bacteria can be grouped or classified in several different ways
- **"True" bacteria** (which include all bacteria that infect man) are members of one kingdom (the **Eubacteria**).
- In addition, a group of organisms often found in extreme environments form a second kingdom (Archaebacteria also called ***Archaea***).
- Morphologically, the two kingdoms of organisms appear similar, especially in the absence of a nucleus, and thus are classified together as prokaryotes.

General Structure of Prokaryotic cell or Bacteria

There are a few basic things that are commonly present in almost all bacteria

- Glycocalyx/Capsule /Slime
- Cell wall (Except *Mycoplasma*)
- Cell membrane
- Cytoplasm (Cytosol)
- Genetic material (Naked DNA/Chromosome/Nucleoid)
- Ribosomes



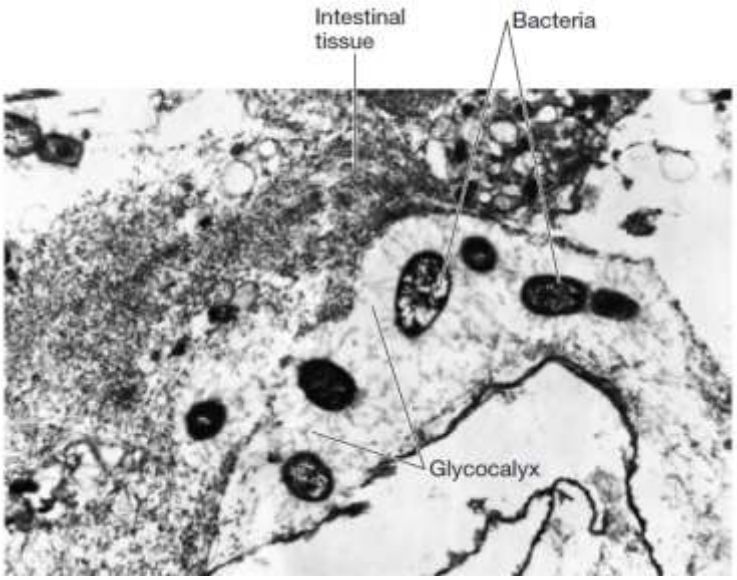
In addition to these basic structures there are a few things that can be seen in some bacteria, like

- Plasmid
- Inclusion Bodies
- Flagella
- Pili and Fimbriae

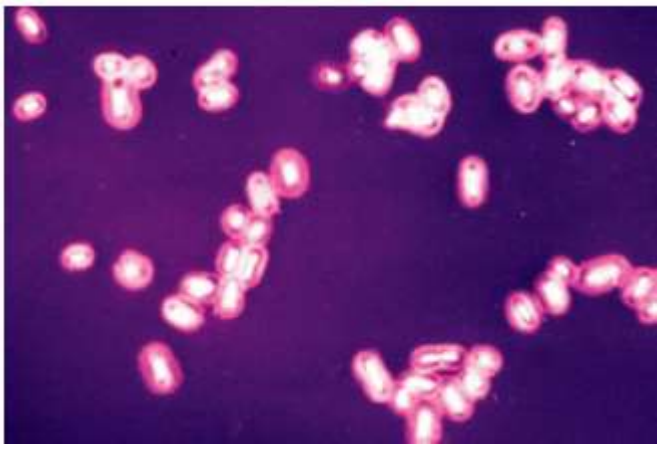
Detailed structure of Prokaryotic cell

GLYCOCALYX/CAPSULE/SLIME

- A **gelatinous polysaccharide** or polypeptide outer covering of certain bacteria is called glycocalyx.
- These are the structures that surround the outside of the cell.
- The glycocalyx is referred to as a **capsule** if it is firmly attached to the cell wall, or as a **slime layer** if loosely attached.
- The chemical nature of bacterial capsules is diverse but majority of them are polysaccharides.



Bacterial Glycocalyx



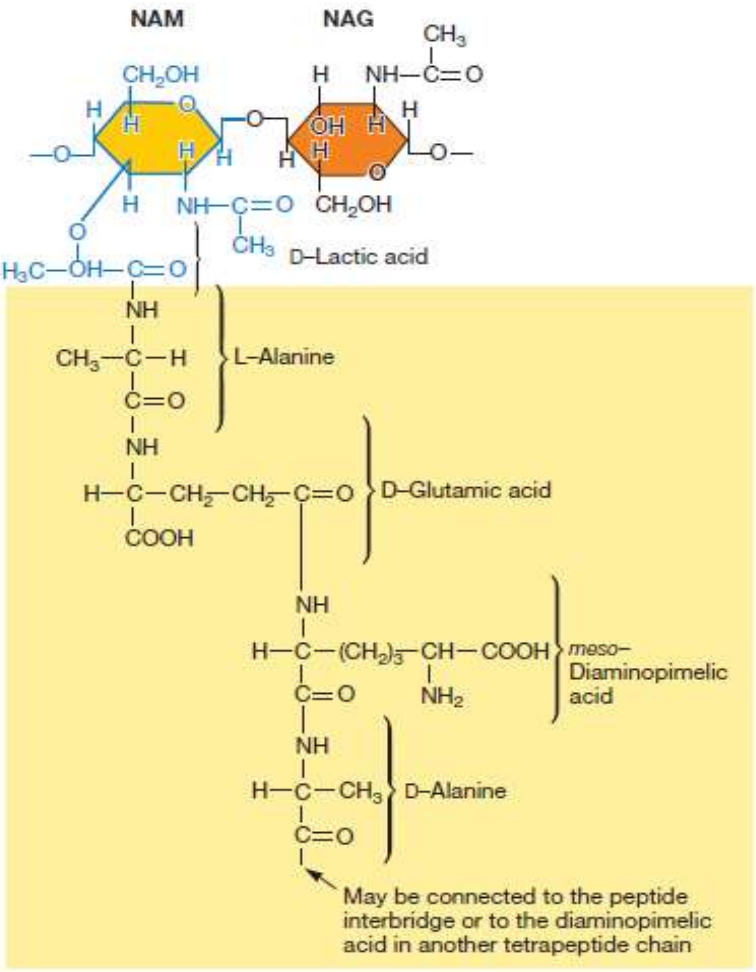
Bacterial Capsules. *Klebsiella pneumoniae* with its capsule stained for observation in the light microscope

Glycocalyx encloses a layer called cell wall.

CELL WALL

- The cell wall is a rigid layer that lies just inside the glycocalyx.

- In gram positive bacteria, the cell wall mainly consists of **peptidoglycan** and **teichoic acid**
- the cell wall in Gram negative bacteria includes **peptidoglycan, lipoprotein, outer membrane** and **lipopolysaccharide layers.**
- Cell wall does not take up any stain and hence are not seen by light microscope.
- Most bacteria have a complex cell wall consisting of peptidoglycan (also called **murein, mucopeptide**). This complex polymer consists of three parts-
    - ✓ A backbone consisting of alternating units of **NAG (N-acetylglucosamine**) and **NAM (N-acetylmuramic acid).**
    - ✓ **Tetrapeptide side chain** attached to NAM
    - ✓ **Peptide cross-bridges**, which are short chains of amino acids that crosslink the backbone.
- Just underneath the cell wall the cell membrane or the plasma membrane is present

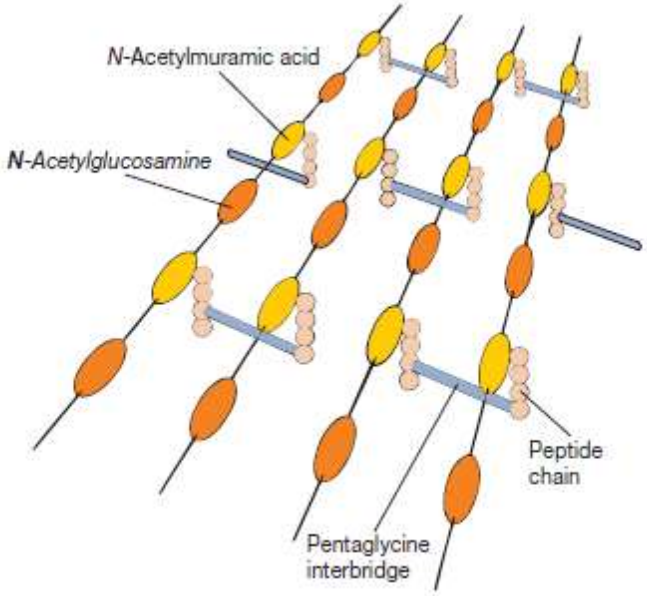


Peptidoglycan subunit composition.

SIGNIFICANCE OF CELL WALL

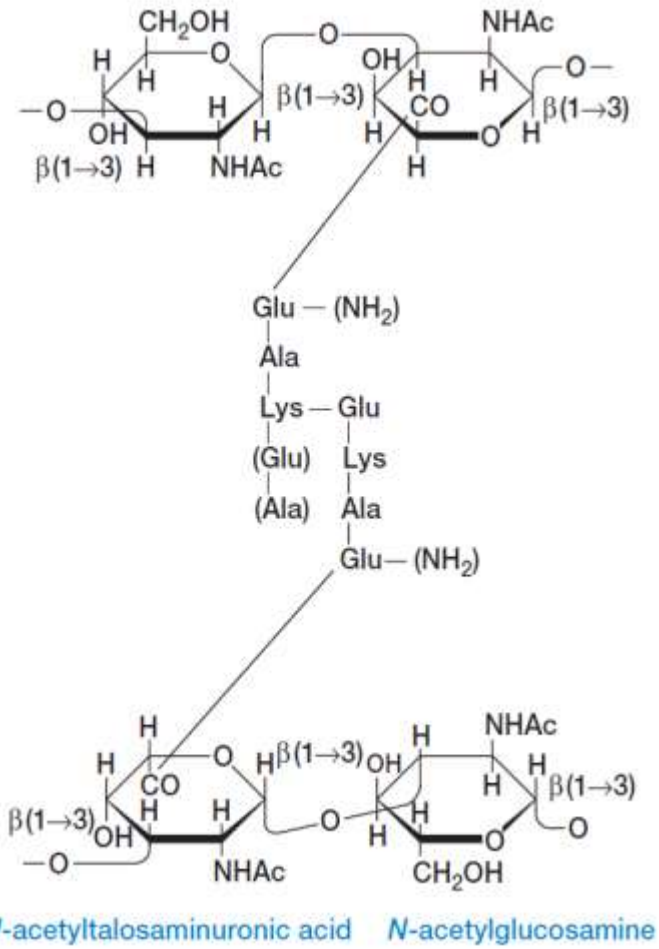
- **Maintains** cell shape, any cell that loses its cell wall, loses its shape as well.
- **Protects** bacteria from osmotic lysis
- Acts as a **barrier**, protects cell contents from external environment
- Determines reactivity to Gram stain, cells become gram negative if they lose cell wall
- Site of action of certain antimicrobial agents (E.g. Penicillin, Cephalosporins)
- Bacteria may attach to surface, produce slime, divide and produce microcolonies within the slime layer and construct a biofilm.

- Confer specific antigenicity to a strain/species that can be exploited to detect and identify an isolate.
- Peptidoglycan is the building block of cell wall and it is formed by **transpeptidation**.



Peptidoglycan structure. A schematic diagram of one model of peptidoglycan. Shown are the polysaccharide chains, tetrapeptide side chains, and peptide interbridges.

Important notes: 1. Peptidoglycan (a heteropolysaccharide) is one of the most important components of bacterial cell wall. 2. It can be digested by the enzyme lysozyme. 3. Its synthesis can be blocked by the antibiotics like Penicillin



The Structure of Peptidoglycan. The amino acids and amino groups in parentheses are not always present. Ac represents the acetyl group.

BIOSYNTHESIS OF PEPTIDOGLYCAN

In order for bacteria to increase their size following **binary fission**, links in the peptidoglycan must be broken, new peptidoglycan monomers must be inserted, and the peptide cross links must be resealed.

The following sequence of events occur:

Step 1. Bacterial enzymes called **autolysins**:

a) Break the **glycosidic bonds** between the peptidoglycan monomers at the point of growth along the existing peptidoglycan and

b) Break the peptide cross-bridges that link the rows of sugars together

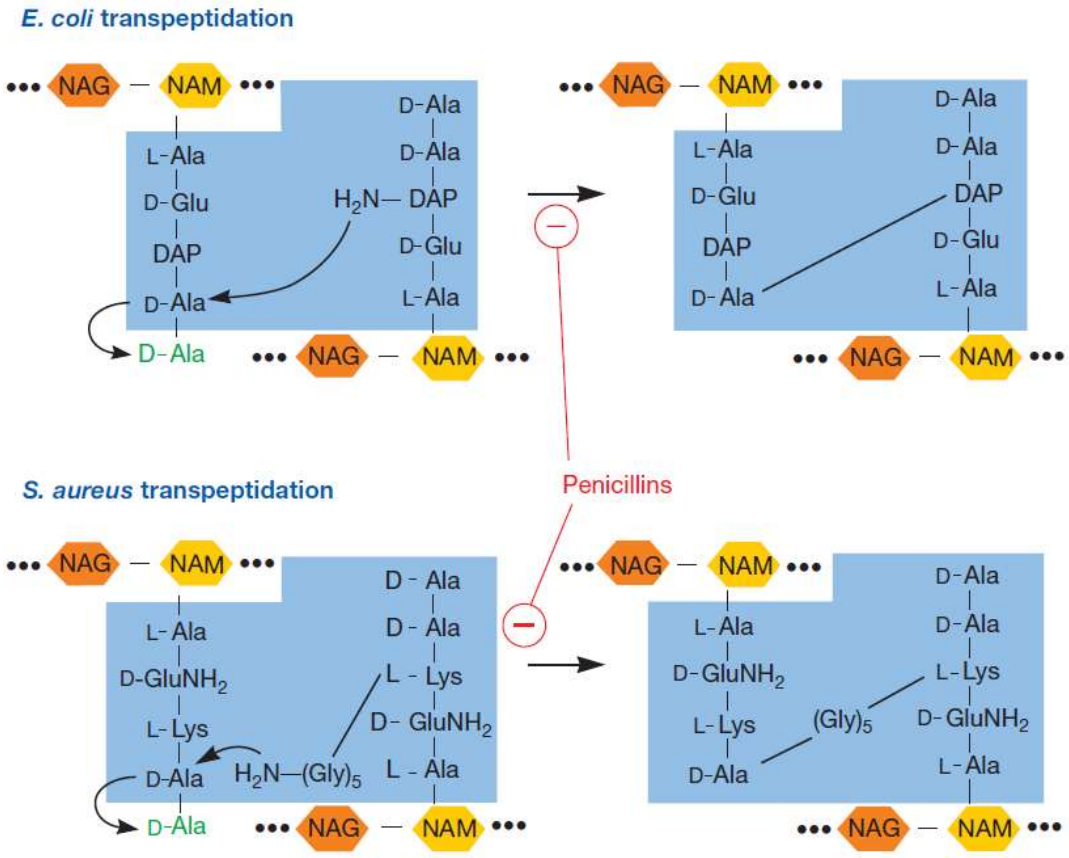
Step 2. The peptidoglycan monomers are synthesized in the cytosol and bind to **bactoprenol**. The bactoprenols transport the peptidoglycan monomers across the cytoplasmic membrane and interacts with **transglycosidases** to insert the monomers into existing peptidoglycan

Step 3. **Transglycosylase** (transglycosidase) enzymes insert and link new peptidoglycan monomers into the breaks in the peptidoglycan

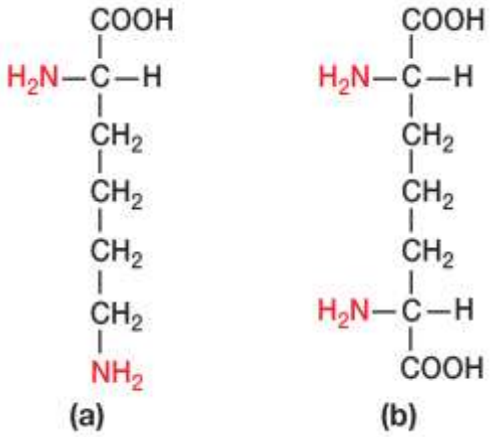
Step 4. Finally, **transpeptidase** enzymes reform the peptide cross-links between the rows and layers of peptidoglycan during transpeptidation to make the wall strong.

TRANSPEPTIDATION

- The final step in peptidoglycan synthesis is transpeptidation, which creates the peptide cross-links between the peptidoglycan chains.
- The transpeptidation reaction occurs in two steps, the first being formation of a covalent enzyme intermediate and the second involving attack of an amine on this intermediate.
- Firstly, polymerization reaction of the newly synthesized disaccharide-peptide units take place, followed by their incorporation into the growing chain.
- This is achieved mainly through the action of proteins, which catalyze the formation of peptide bonds of the peptidoglycan.
- Then the bond between two adjacent amino acids (d-Alanine), called D-Ala-D-Ala bond of one peptide chain is cleaved and an enzyme-substrate intermediate is formed, with the release of the terminal amino acid(d-Alanine) of the peptide chain.
- The cleavage reaction provides the energy necessary for the reaction, which occurs outside the cytoplasmic membrane in the absence of energy donors such as ATP.
- A second step involves the transfer of the peptidyl moiety to an acceptor.
- This acceptor is the non-alpha amino group of the dibasic amino acids in a second stem peptide (in bacteria with direct cross-linking) or the last amino acid of the cross bridge.
- The reaction results in the formation of a new peptide bond between the amino acid (d-alanine) of a donor peptide and an amino group of the cross bridge of an acceptor peptide.

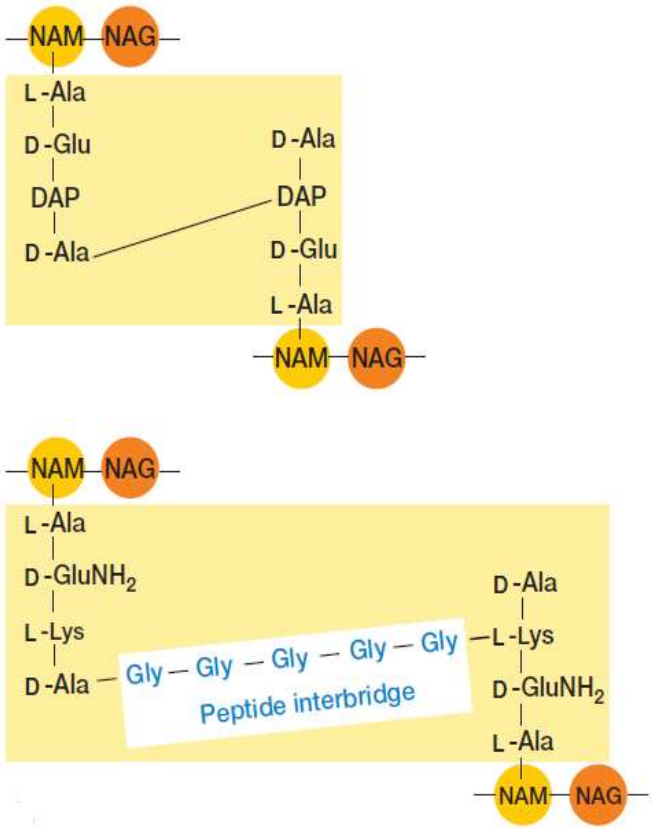


Transpeptidation. The transpeptidation reactions in the formation of the peptidoglycans of *Escherichia coli* and *Staphylococcus aureus.* In *Escherichia coli*, the terminal D-alanine is cleaved from the pentapeptides to form a tetrapeptides. This provides the energy to bond the D-alanine of one tetrapeptide to the diaminopimelic acid of another tetrapeptide. In the case of *Staphylococcus aureus*, the terminal D-alanine is cleaved from the pentapeptides to form a tetrapeptides. This provides the energy to bond a pentaglycine bridge (5 molecules of the amino acid glycine) from the D-alanine of one tetrapeptide to the L-lysine of another.



Dibasic-aminoacids Present in Peptidoglycan.

(a) L-Lysine. (b) meso-Diaminopimelic acid.

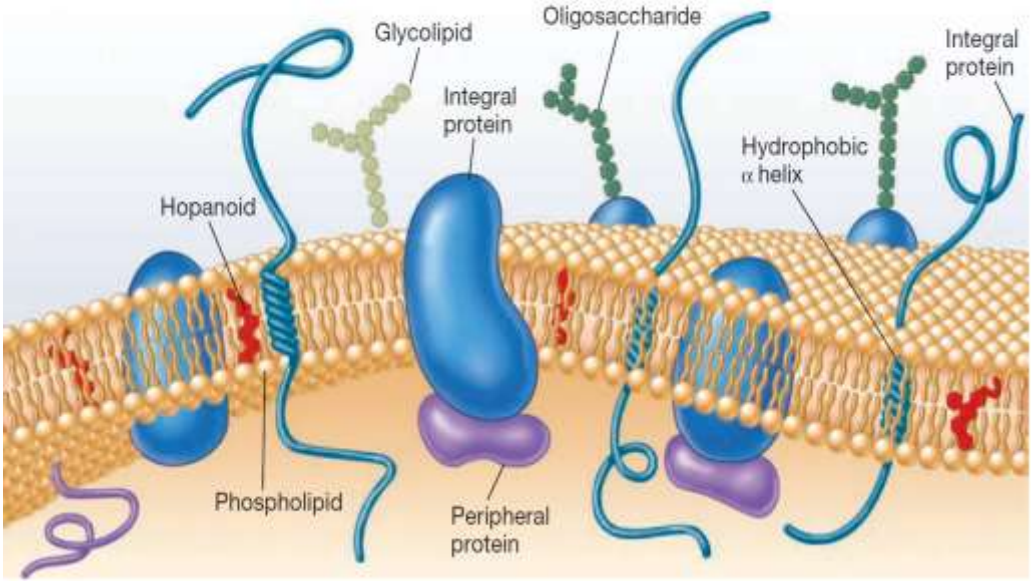


Peptidoglycan Crosslinks

Just underneath the cell wall, the cell membrane or the plasma membrane is present

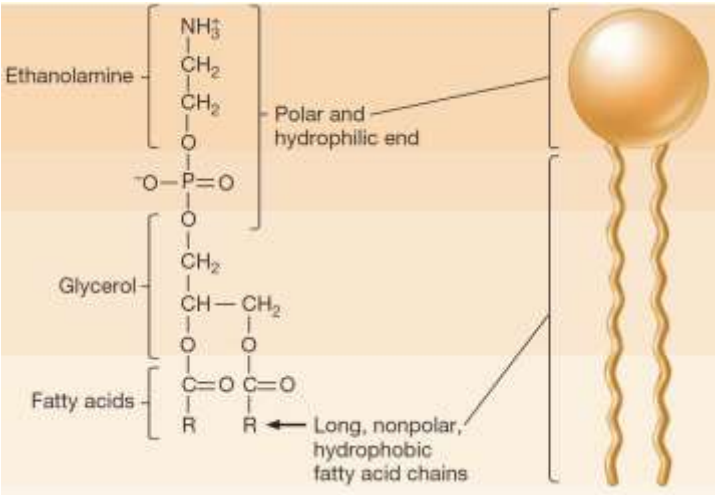
PLASMA MEMBRANE

- The **cell membrane** or plasma membrane is primarily composed of **phospholipids** and **proteins**.
- It measures approximately 5-10 nm in thickness.
- It lies below the **peptidoglycan** layer of the cell wall and encloses the cytoplasm.
- The arrangement of proteins and lipids to form a membrane is called the **fluid mosaic model**.
- The membranes of bacteria (except ***Mycoplasma***) do not contain **sterols.**
- It is a **phospholipid bilayer** with **polar heads** on either side of the membrane.
- **Hydrophobic tails** are oriented to the interior of the membrane.
- Specialized structures called **mesosomes** or **chondroids** are formed from the **convoluted invaginations** of cytoplasmic membrane.
- It encapsulates a semi- liquid called cytoplasm.



Bacterial Plasma Membrane Structure. This diagram of the fluid mosaic model of bacterial membrane structure shows the integral proteins (blue) floating in a lipid bilayer. Peripheral proteins (purple) are associated loosely with the inner membrane surface. Small

spheres represent the hydrophilic ends of membrane phospholipids and wiggly tails, the hydrophobic fatty acid chains. Other membrane lipids such as hopanoids (red) may be present. For the sake of clarity, phospholipids are shown in proportionately much larger size than in real membranes.



The Structure of a Polar Membrane Lipid. Phosphatidylethanolamine, an amphipathic phospholipid often found in bacterial membranes. The R groups are long, nonpolar fatty acid chains.

FUNCTIONS

- A **selectively permeable barrier**: only few substances can pass through the membrane depending upon pore sizes and the hydrophobic nature of the membrane.
- **Integral (transmembrane) proteins** form channels and act as **carriers**.
- **Peripheral proteins** can act as **receptors** and as **enzymes** for metabolic reactions.
- **Electron transport and oxidative phosphorylation**: cytochromes and dehydrogenases of respiratory chain are located in the cell membrane.
- **Excretion** of hydrolytic enzymes.
- Site of attachment of the chromosome.
- Bear receptors and proteins of sensory transduction system.
- The plasma membrane encapsulates a gel-like fluid, called cytoplasm.

Important notes: 1. Plasma membrane is made of phospholipid bilayer. 2. Phospholipids are amphipathic molecules 3. It is important to understand that the parts of the proteins that are embedded in the membrane are made of hydrophobic amino acids (Try to remember the list of all hydrophobic amino acids)

Membrane Transport Mechanisms

All cells acquire the molecules and ions they need from their surrounding extracellular fluid (ECF). There is an unceasing traffic of molecules and ions in and out of the cell through its plasma membrane, for example, glucose, Na+, Ca2+

In eukaryotic cells, there is also transport in and out of membrane-bounded intracellular compartments such as the nucleus, endoplasmic reticulum, and mitochondria, for example, proteins, mRNA, Ca2+, ATP.

Two Problems To Be Considered

1. Relative concentrations

Molecules and ions move spontaneously down their concentration gradient (i.e. , from a region of higher to a region of lower concentration) by diffusion. Molecules and ions can be moved against their concentration gradient, but this process, called active transport, requires the expenditure of energy (usually from ATP).

2. Lipid bilayers are impermeable to most essential molecules and ions

The lipid bilayer is permeable to water molecules and a few other small, uncharged, molecules like oxygen (O2) and carbon dioxide (CO2). These diffuse freely in and out of the cell. The diffusion of water through the plasma membrane is of such importance to the cell that it is given a special name: osmosis. Lipid bilayers are **not** permeable to **ions** such as

- K^+, Na^+, Ca^{2+} (called **cations** because when subjected to an electric field they migrate toward the cathode [the negatively-charged electrode])
- Cl^-, HCO_3^- (called **anions** because they migrate toward the anode [the positively-charged electrode])
- small hydrophilic molecules like glucose
- macromolecules like proteins and RNA

This page will examine how ions and small molecules are transported across cell membranes. The transport of macromolecules through membranes is described in Endocytosis.

SOLVING THESE PROBLEMS

Mechanisms by which cells solve the problem of transporting ions and small molecules across their membranes:

- Facilitated diffusion
    - Transmembrane proteins create a water-filled pore through which ions and some small hydrophilic molecules can pass by diffusion. The channels can be opened (or closed) according to the needs of the cell.
- Active transport
    - Transmembrane proteins, called transporters, use the energy of ATP to force ions or small molecules through the membrane against their concentration gradient.

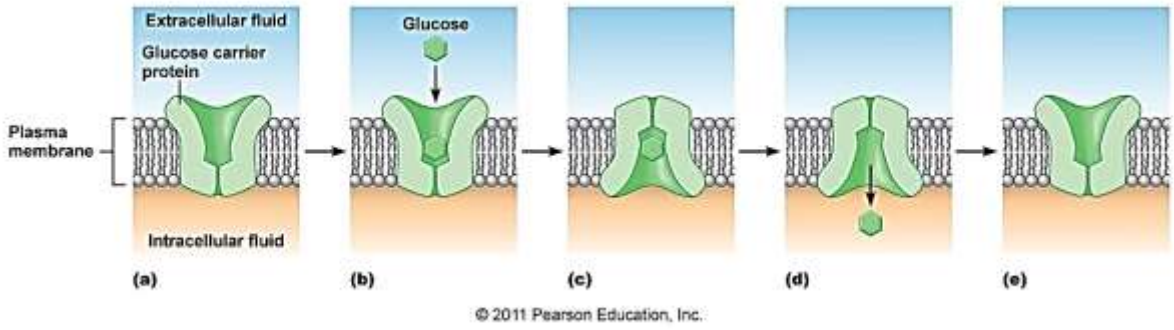
FACILITATED DIFFUSION OF IONS

Facilitated diffusion of ions takes place through proteins, or assemblies of proteins, embedded in the plasma membrane. These transmembrane proteins form a water-filled channel through which the ion can pass **down** its concentration gradient.

The transmembrane channels that permit facilitated diffusion can be opened or closed. They are said to be **"gated"**.

Some types of gated ion channels:

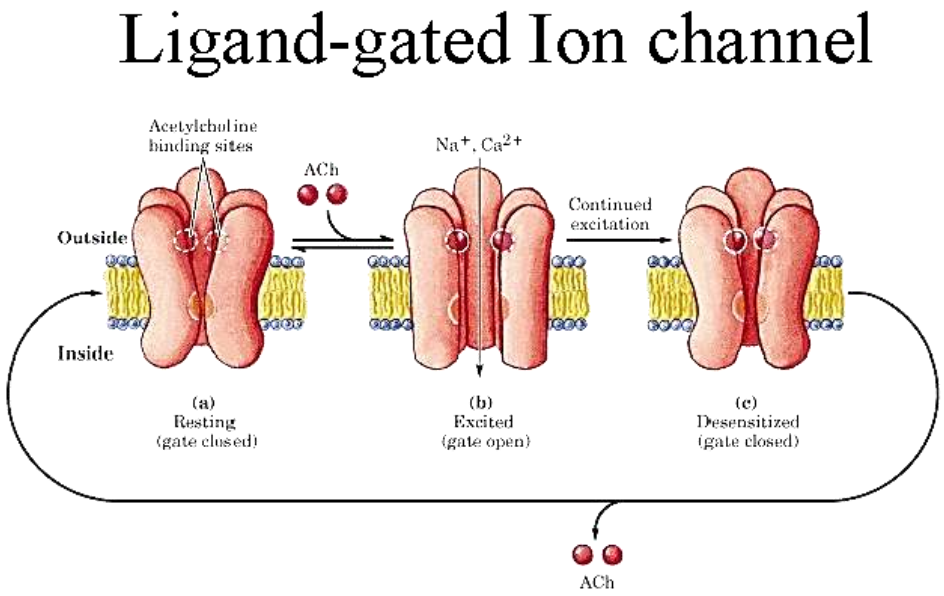
- ligand-gated
- mechanically-gated
- voltage-gated
- light-gated



© 2011 Pearson Education, Inc.

LIGAND-GATED ION CHANNELS

Many ion channels open or close in response to binding a small signaling molecule or **"ligand"**. Some ion channels are gated by extracellular ligands; some by intracellular ligands. In both cases, the ligand is **not** the substance that is transported when the channel opens.



EXTERNAL LIGANDS

External ligands (shown here in green) bind to a site on the extracellular side of the channel. Some example are:

- Acetylcholine (ACh). The binding of the neurotransmitter acetylcholine at certain synapses opens channels that admit Na+ and initiate a nerve impulse or muscle contraction.
- Gamma amino butyric acid (GABA). Binding of GABA at certain synapses - designated GABAA - in the central nervous system admits Cl- ions into the cell and inhibits the creation of a nerve impulse.

INTERNAL LIGANDS

Internal ligands bind to a site on the channel protein exposed to the cytosol. Some examples,

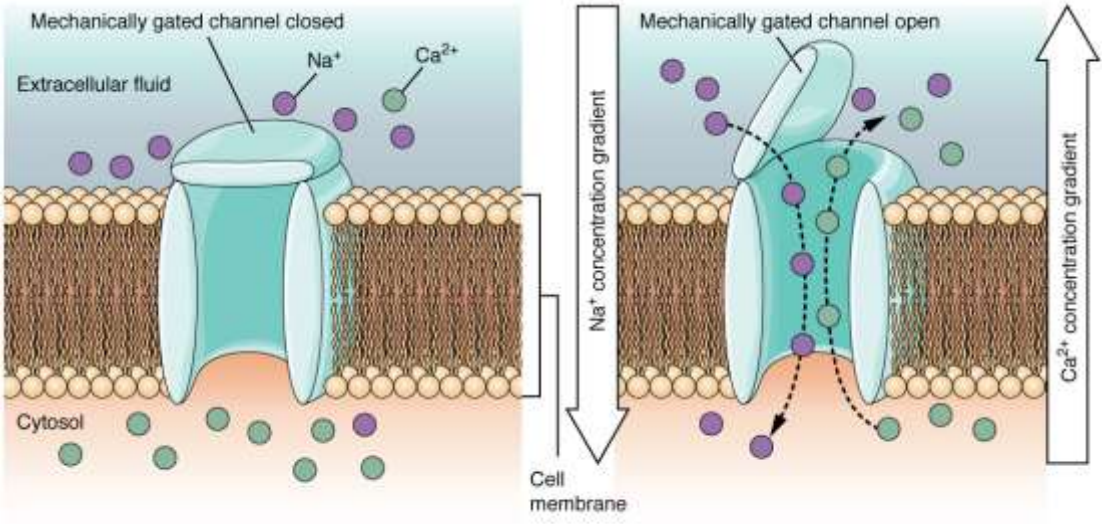
- "Second messengers", like cyclic AMP (cAMP) and cyclic GMP (cGMP), regulate channels involved in the initiation of impulses in neurons responding to odors and light respectively.
- ATP is needed to open the channel that allows chloride (Cl-) and bicarbonate (HCO3-) ions out of the cell. This channel is defective in patients with cystic fibrosis. Although the energy liberated by the hydrolysis of ATP is needed to open the channel, this is not an example of

active transport; the ions diffuse through the open channel following their concentration gradient.

MECHANICALLY-GATED ION CHANNELS

Examples:

- Sound waves bending the cilia-like projections on the hair cells of the inner ear open up ion channels leading to the creation of nerve impulses that the brain interprets as sound.
- Mechanical deformation of the cells of stretch receptors opens ion channels leading to the creation of nerve impulses.

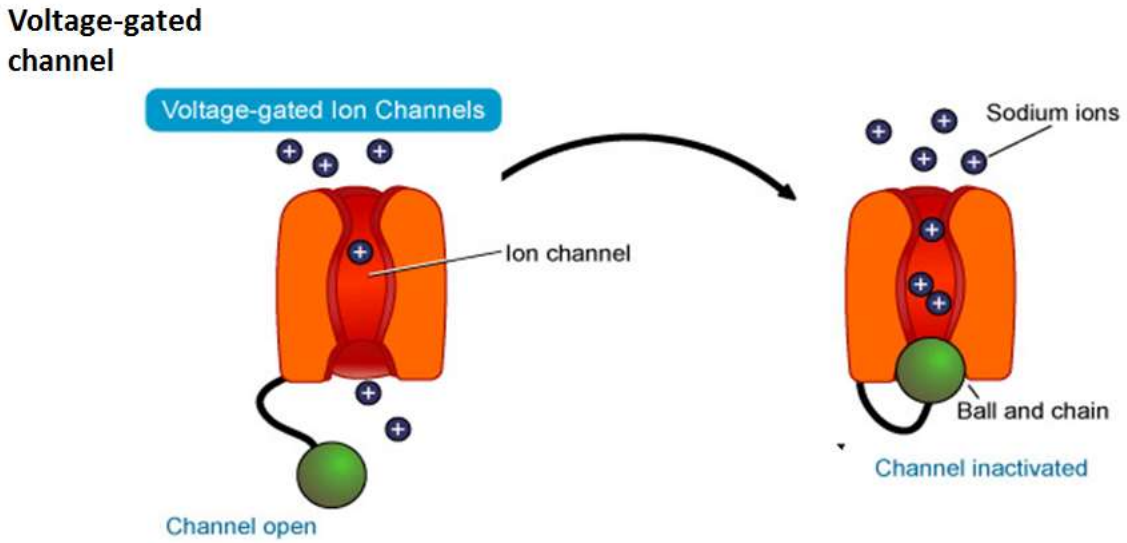


VOLTAGE-GATED ION CHANNELS

In so-called "excitable" cells like neurons and muscle cells, some channels open or close in response to changes in the charge (measured in volts) across the plasma membrane.

Example: As an impulse passes down a neuron, the reduction in the voltage opens sodium channels in the adjacent portion of the membrane. This allows the influx of Na+ into the neuron and thus the continuation of the nerve impulse.

Some 7000 sodium ions pass through each channel during the brief period (about 1 millisecond) that it remains open. This was learned by use of the patch clamp technique.



THE PATCH CLAMP TECHNIQUE

The properties of ion channels can be studied by means of the patch clamp technique.

- A very fine pipette (with an opening of about 0.5 µm) is pressed against the plasma membrane of either an intact cell; or
- The plasma membrane can be pulled away from the cell and the preparation placed in a test solution of desired composition.
- Current flow through a single ion channel can then be measured.

Such measurements reveal that each channel is either fully open or fully closed; that is, facilitated diffusion through a single channel is "all-or-none".

This technique has provided so much valuable information about ion channels that its inventors, Erwin Neher and Bert Sakmann, were awarded a Nobel Prize in 1991.

FACILITATED DIFFUSION OF MOLECULES

Some small, hydrophilic organic molecules, like sugars, can pass through cell membranes by facilitated diffusion.

Once again, the process requires transmembrane proteins. In some cases, these - like ion channels - form water-filled pores that enable the molecule to pass in (or out) of the membrane following its concentration gradient.

Example: Maltoporin. This homotrimer in the outer membrane of E. coli forms pores that allow the disaccharide maltose and a few related molecules to diffuse into the cell.

Another example: The plasma membrane of human red blood cells contains transmembrane proteins that permit the diffusion of glucose from the blood into the cell.

Note that in all cases of facilitated diffusion through channels, the channels are selective; that is, the structure of the protein admits only certain types of molecules through.

Whether all cases of facilitated diffusion of small molecules use channels is yet to be proven. Perhaps some molecules are passed through the membrane by a conformational change in the shape of the transmembrane protein when it binds the molecule to be transported.

ACTIVE TRANSPORT

Active transport is the pumping of molecules or ions through a membrane **against** their concentration gradient. It requires:

- a transmembrane protein (usually a complex of them) called a transporter and
- energy. The source of this energy is ATP.

The energy of ATP may be used directly or indirectly.

- **Direct Active Transport.** Some transporters bind ATP directly and use the energy of its hydrolysis to drive active transport.
- **Indirect Active Transport.** Other transporters use the energy already stored in the gradient of a directly-pumped **ion**. Direct active transport of the ion establishes a concentration gradient. When this is relieved by facilitated diffusion, the energy released can be harnessed to the pumping of some other ion or molecule.

DIRECT ACTIVE TRANSPORT

1. THE NA+/K+ ATPASE (PUMP)

The cytosol of animal cells contains a concentration of potassium ions (K+) as much as 20 times higher than that in the extracellular fluid. Conversely, the extracellular fluid contains a concentration of sodium ions (Na+) as much as 10 times greater than that within the cell.

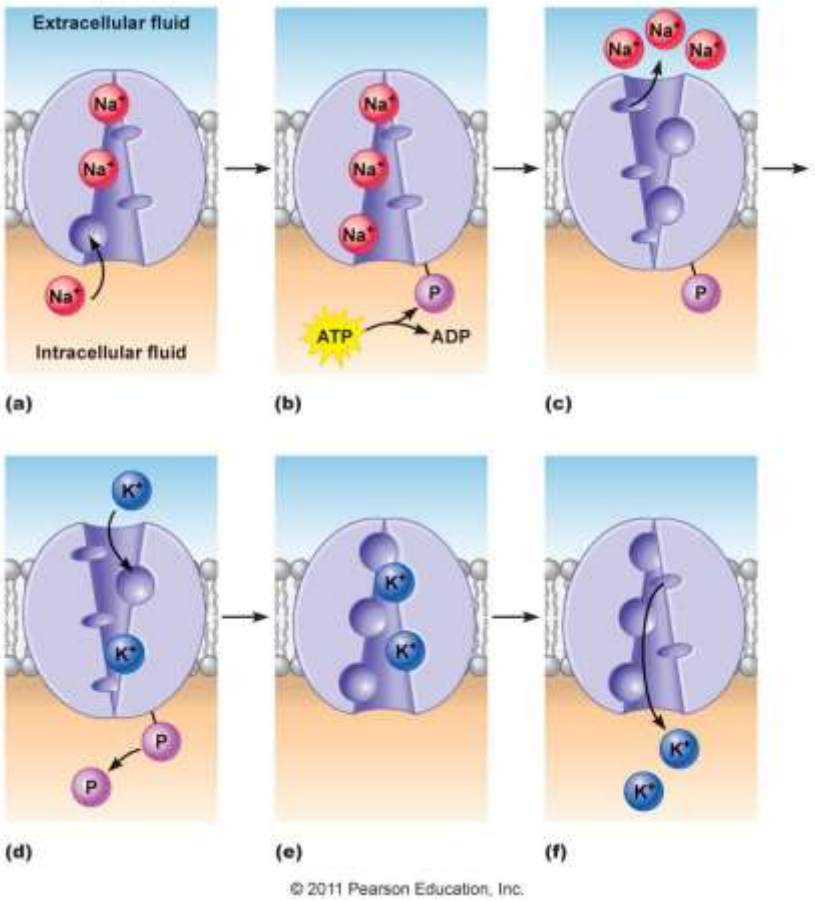
These concentration gradients are established by the active transport of both ions. And, in fact, the same transporter, called the Na+/K+ ATPase, does both jobs. It uses the energy from the hydrolysis of ATP to

- actively transport 3 Na^+ ions out of the cell
- for each 2 K^+ ions pumped into the cell.

This accomplishes several vital functions:

- It helps establish a net charge across the plasma membrane with the interior of the cell being negatively charged with respect to the exterior. This resting potential prepares nerve and muscle cells for the propagation of action potentials leading to nerve impulses and muscle contraction.
- The accumulation of sodium ions outside of the cell draws water out of the cell and thus enables it to maintain osmotic balance (otherwise it would swell and burst from the inward diffusion of water).
- The gradient of sodium ions is harnessed to provide the energy to run several types of indirect pumps.

The crucial roles of the Na+/K+ ATPase are reflected in the fact that almost one-third of all the energy generated by the mitochondria in animal cells is used just to run this pump.



© 2011 Pearson Education, Inc.

2. The H+/K+ ATPase

The parietal cells of your stomach use this pump to secrete gastric juice. These cells transport protons (H+) from a concentration of about 4 x 10-8 M within the cell to a concentration of about 0. 15 M in the gastric juice (giving it a pH close to 1). Small wonder that parietal cells are stuffed with mitochondria and uses huge amounts of ATP as they carry out this three-million-fold concentration of protons.

3. The Ca2+ ATPases

A Ca^{2+} ATPase is located in the **plasma membrane** of all eukaryotic cells. It uses the energy provided by one molecule of ATP to pump one Ca^{2+} ion out of the cell. The activity of these pumps helps to maintain the ~20, 000-fold concentration gradient of Ca^{2+} between the cytosol (~ 100 nM) and the ECF (~ 20 mM).

In resting skeletal muscle, there is a much higher concentration of calcium ions (Ca^{2+}) in the sarcoplasmic reticulum than in the cytosol. Activation of the muscle fiber allows some of this Ca^{2+} to pass by facilitated diffusion into the cytosol where it triggers contraction.

After contraction, this Ca^{2+} is pumped back into the sarcoplasmic reticulum. This is done by another Ca^{2+} ATPase that uses the energy from each molecule of ATP to pump 2 Ca^{2+} ions.

Pumps **1-3** are designated **P-type ion transporters** because they use the same basic mechanism: a conformational change in the proteins as they are reversibly phosphorylated by ATP. And all three pumps can be made to run backward. That is, if the pumped ions are allowed to diffuse back through the membrane complex, ATP can be synthesized from ADP and inorganic phosphate.

4. ABC Transporters

ABC ("ATP-Binding Cassette") transporters are transmembrane proteins that

- expose a ligand-binding domain at one surface and a
- ATP-binding domain at the other surface.

The ligand-binding domain is usually restricted to a single type of molecule.

The ATP bound to its domain provides the energy to pump the ligand across the membrane.

The human genome contains 48 genes for ABC transporters. Some examples:

- CFTR - the cystic fibrosis transmembrane conductance regulator
- TAP, the transporter associated with antigen processing.
- the transporter that liver cells use to pump the salts of bile acids out into the bile.

- ABC transporters that pump chemotherapeutic drugs out of cancer cells thus reducing their effectiveness.

ABC transporters must have evolved early in the history of life. The ATP-binding domains in archaea, eubacteria, and eukaryotes all share a homologous structure, the ATP-binding "cassette".

INDIRECT ACTIVE TRANSPORT

Indirect active transport uses the downhill flow of an ion to pump some other molecule or ion against its gradient. The driving ion is usually sodium (Na^+) with its gradient established by the Na^+/K^+ ATPase.

SYMPORT PUMPS

In this type of indirect active transport, the driving ion (Na^+) and the pumped molecule pass through the membrane pump in the **same** direction. Some examples are:

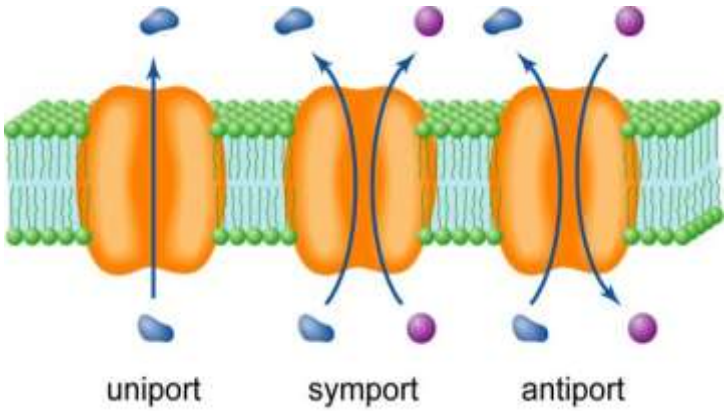
- The Na+/glucose transporter. This transmembrane protein allows sodium ions and glucose to enter the cell together. The sodium ions flow down their concentration gradient while the glucose molecules are pumped up theirs. Later the sodium is pumped back out of the cell by the Na+/K+ ATPase.
- The Na+/glucose transporter is used to actively transport glucose out of the intestine and also out of the kidney tubules and back into the blood.
- All the amino acids can be actively transported, for example out of the kidney tubules and into the blood, by sodium-driven symport pumps.
- Sodium-driven symport pumps also return neurotransmitters to the presynaptic neuron .
- The Na+/iodide transporter. This symporter pumps iodide ions into the cells of the thyroid gland (for the manufacture of thyroxine) and also into the cells of the mammary gland (to supply the baby's need for iodide).
- The permease encoded by the lac operon of E. coli that transports lactose into the cell.

ANTIPORT PUMPS

In anti-port pumps, the driving ion (again, usually sodium) diffuses through the pump in one direction providing the energy for the active transport of some other molecule or ion in the opposite direction. Example: Ca2+ ions are pumped out of cells by sodium-driven anti-port pumps.

Antiport pumps in the vacuole of some plants harness the outward facilitated diffusion of protons (themselves pumped into the vacuole by aH+ ATPase).

- to the active inward transport of sodium ions. This sodium/proton anti-\port pump enables the plant to sequester sodium ions in its vacuole. Transgenic tomato plants that overexpress this sodium/proton antiport pump are able to thrive in saline soils too salty for conventional tomatoes.
- to the active inward transport of nitrate ions (NO_3^-).



Important notes: 1. Passive transport does not require energy. 2. Active transport needs energy (ATP). It is important to understand and remember the mechanisms of some of the transporters like the Sodium-Potassium ATPase

MESOSOMES

- They are organelles formed as a result of invagination of the bacterial plasma membrane.
- These localized infoldings are unique to the bacteria
- They play a very important role in the respiration of bacteria.
- It also helps in binding the bacterial DNA during the cell division

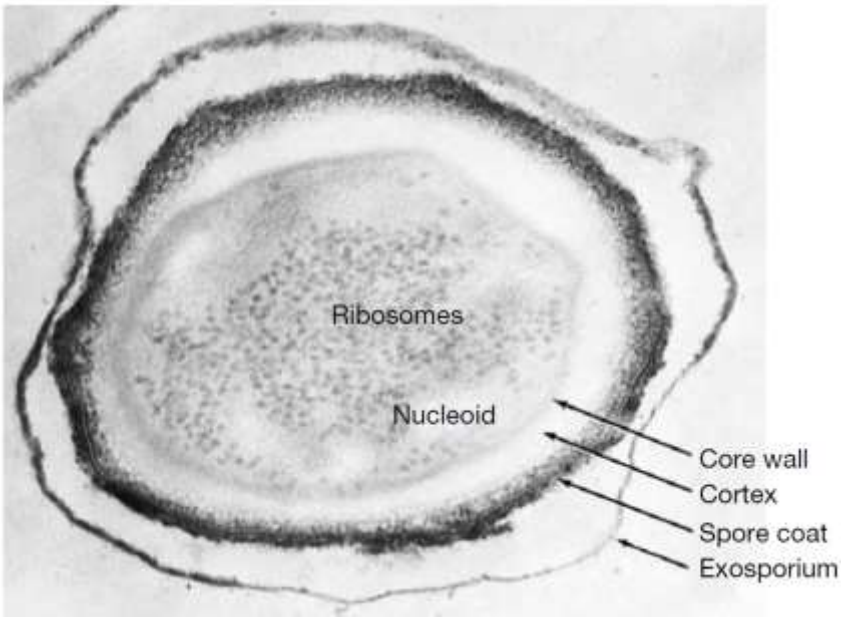
CYTOPLASM

- The cytoplasm or protoplasm is the portion of the cell that lies within the cytoplasmic membrane.
- It is **gel-like** in consistency and includes the prokaryotic chromosome and ribosomes.
- Constituents of cytoplasm include proteins, vitamins, ions, nucleic acids, amino acids, carbohydrates, fatty acids.
- The cytoplasm does not exhibit any internal mobility (**cytoplasmic streaming**).
- It also contains floating genetic material

ENDOSPORES

- In poor growth conditions some bacteria such as ***Bacillus*** and ***Clostridium*** produce **resistant** survival forms termed endospores. This process is known as **sporulation.**
- Bacterial spores are endospores in contrast to fungal spores, which are called **exospores**.
- Unlike the spores of fungi, bacterial spores do not serve reproductive function.
- They are resistant to **extreme** environmental conditions such as high temperatures, dryness, toxic chemicals (disinfectants, antibiotics), and UV radiation.
- Once the endospore is formed, the vegetative portion of the bacterium is degraded and the dormant endospore is released.
- The endospore is able to survive for long periods of time until environmental conditions again become favorable for growth.
- The endospore then germinates, producing a single vegetative bacterium.
- Only some spores can be killed by **sterilization** methods such as autoclave and hot air oven.

- Some chemical **disinfectants** such as formaldehyde and ethylene oxide can also kill spores.



Endospore Structure

SIGNIFICANCE OF ENDOSPORES

- Since they are **resistant** forms of bacteria, they can survive **unfavourable** conditions for long period.
- Since spores occur in **soil**, wounds contaminated by soil can lead to infections like gangrene or tetanus.
- Since spores survive ordinary disinfection, they may contaminate surgical wounds.
- Since spores are everywhere, they may contaminate bacterial culture media.
- Since they are highly heat resistant, they can be used to monitor the efficacy of sterilization process in autoclave (*Bacillus stearothermophilus*) and hot air oven (*Clostridium tetani* or *niger*).

GAS VESICLES

- Some of the planktonic bacteria contain spindle-shaped structures called the gas vesicles.
- These vesicles are hollow and are made of protein.
- They usually have cylindrical shape with conical end caps

- These vesicles decrease the overall cell density of the cells and provides the buoyancy necessary for these cells to keep themselves in the upper reaches of the water column
- This positive buoyancy ensures that the bacteria stay on the upper layers of the water and there by continue to make their food through photosynthesis
- These gas vesicles are found in 5 phyla of bacteria and in some archaebacteria as well.

NUCLEOID (GENETIC MATERIAL)

- The genetic material of prokaryotes is not surrounded by a nuclear membrane. This is called **nucleoid**
- The nucleoid or bacterial chromosome comprises a closed circle of double stranded DNA, many times the length of the cell and highly folded and compacted.
- The common laboratory bacterium *Escherichia coli* is around 3–4 μm in length, but contains a DNA molecule some 1400 μm in length.
- **Eukaryotes** have one or more **linear** chromosomes that are located in the **nucleus.**
- **Prokaryotes** have a single **circular** chromosome located in the **cytoplasm**
- The DNA may be associated with certain bacterial proteins, but these are not the same as the **histones** found in eukaryotic chromosomes.
- Some bacteria contain additional DNA in the form of small, self-replicating extrachromosomal elements called **plasmids**.

Important notes: 1. Average molecular weight of a DNA base pair (sodium salt) = 650 daltons (1 dalton equals the mass of a single hydrogen atom, or 1.67 x 10^{-24} grams).

PLASMID

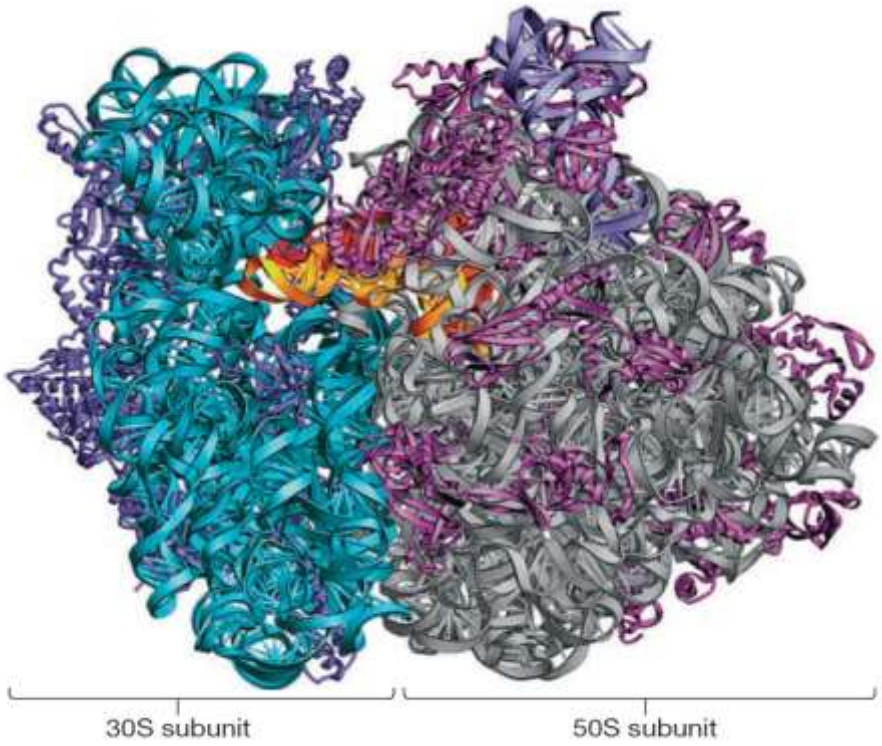
- In addition to the nucleoid, certain bacteria also contain an extrachromosomal DNA called the plasmids.
- Plasmids are small, **extrachromosomal** and double-stranded DNA molecules that can exist **independently** of the chromosome.
- Plasmids can be **circular** as well as **linear**, but most common being circular.

- Plasmids may be classified in terms of their mode of existence, spread, and function.
- **Single-copy** plasmids produce only one copy per host cell.
- **Multicopy** plasmids may be present at concentrations of 40 or more per cell.
- Some plasmids are able to **integrate** into the chromosome and are called **episomes.**
- **Conjugative** plasmids cause the generation of hairlike structures called **pili** and can transfer themselves to other bacteria during **conjugation**.
- Example of conjugative plasmid is the **F factor** (**fertility factor or F plasmid**) of *E. coli*
- **Resistance** factors (**R factors, R plasmids**) can confer **antibiotic resistance** on the cells that contain them.
- R factors contain genes that code for enzymes capable of destroying or modifying antibiotics.
- They are not integrated into the host chromosome.
- **Col plasmids** contain genes for the synthesis of bacteriocins known as **colicins**, which are directed against E. coli.
- Some Col plasmids are conjugative and carry resistance genes.
- For example, the bacteriocin genes of *Pseudomonas aeruginosa*, which code for proteins called pyocins, are located on the chromosome.
- **Virulence** plasmids make their hosts more **pathogenic.**
- For example, enterotoxigenic strains of *E. coli* cause traveler's diarrhea because they contain a plasmid that codes for an enterotoxin.
- **Metabolic** plasmids carry genes for enzymes that **degrade** substances such as aromatic compounds (toluene), pesticides (2,4-dichlorophenoxyacetic acid), and sugars (lactose).

RIBOSOMES

- Ribosomes are composed of a complex of **protein and RNA**.
- They are the site of **protein synthesis** in the cell
- The ribosomes of prokaryotic cells are smaller and lighter than eukaryotic cells.
- Ribosomes are measured in **Svedberg units** (S), a function of their size and shape, i.e. determined by their rate of sedimentation in a centrifuge

- Prokaryotic ribosomes are **70S**, while those of eukaryotes are 80S.
- All ribosomes comprise two unequal subunits (in prokaryotes, these are 50S and 30S, in eukaryotes - 60S and 40S).
- Each subunit contains its own RNA and a number of proteins
- A **polyribosome (polysome**) is a chain of ribosomes attached to the same molecule of mRNA
- Many ribosomes are simultaneously attached to a single mRNA molecule, forming a threadlike polysome.



Prokaryotic Ribosome. The two subunits of a bacterial ribosome are shown. The 50S subunit includes 23S rRNA (gray) and 5S rRNA (light blue), while 16S rRNA (cyan) is found in the 30S subunit.

FLAGELLA

- Some bacteria are **motile** (can move) and some are not.
- Almost all motile bacteria possess flagella as the organ of **locomotion**.

- Such bacteria tend to move towards or away from the source of **stimulus**.
- These stimuli can be chemicals (**chemotaxis**), light (**phototaxis**), air (**aerotaxis**) or magnetism (**magnetotaxis**).

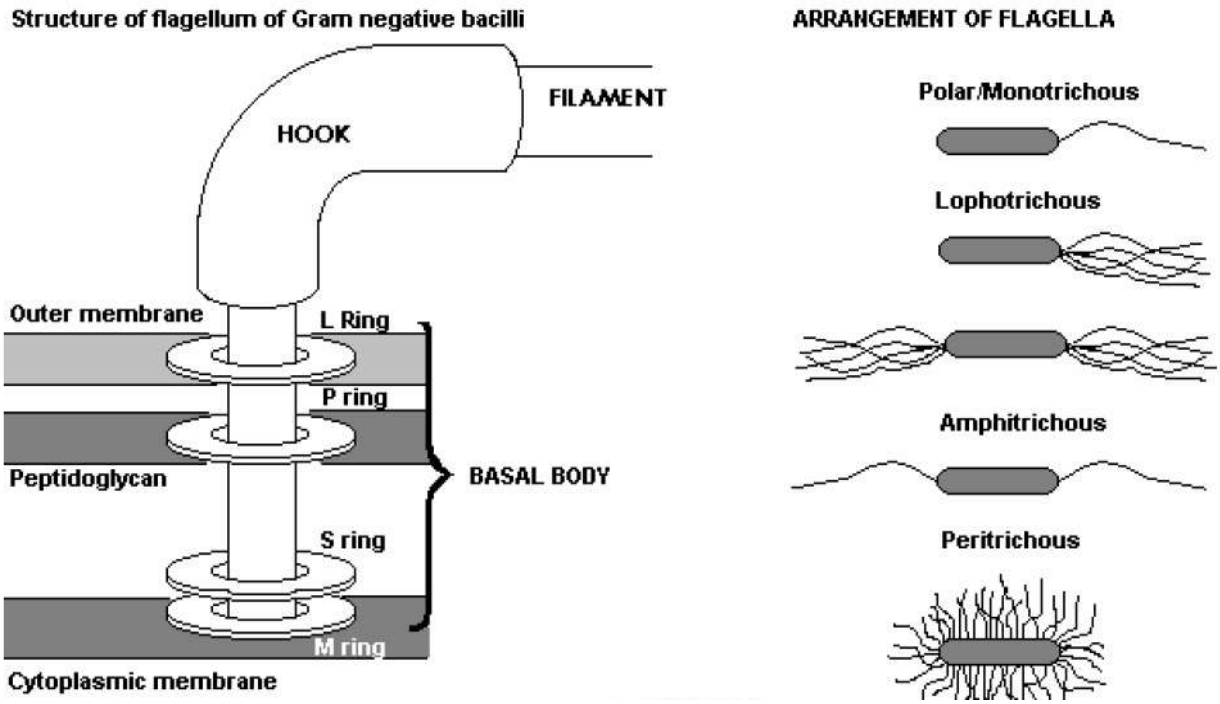
STRUCTURE

- Prokaryotic flagella are much thinner than eukaryotic flagella and they lack the typical arrangement of **microtubules**.
- They are approximately 3-20μm long and end in a square tip.
- Since flagella are very thin (20-30 nm in diameter), and cannot be seen by a normal light microscope.
- The bacterial flagellum is a **non-contractile**, composed of single protein subunit called **flagellin**.
- It is anchored to the bacterial cytoplasmic membrane and cell well by means of disk like structures.
- A flagellum comprises of three parts**, filament, hook** and **basal body**.
- The flagellum is attached to the cell body by hook and basal body.
- While the hook and basal body are embedded in the cell envelope, the filament is free.
- If a flagellum is cut off it will **regenerate** until reaches a maximum length.
- The basal body bears a set of **rings**, one pair in gram positive bacteria and two pairs in gram negative bacteria.
- The rings named **S and M** are common to both gram positive and negative bacteria, the rings names **P and L** are found only in gram negative bacteria.
- Rings in the basal body **rotate** relative to each other causing the flagella to turn like a propeller.
- The energy to drive the basal body is obtained from the **proton motive force**.
- The numbers of flagella, as well as their location on the cell surface are characteristic of a species.

Flagella arrangements are

- **Monotrichous** - a single flagellum at one pole (also called polar flagellum) E.g. *Vibrio cholerae*

- **Amphitrichous** - single flagellum at both poles. Eg. Spirilla
- **Lophotrichous** - two or more flagella at one or both poles of the cell E.g. *Spirillum undula*
- **Peritrichous** - completely surrounded by flagella E.g. *E.coli.*



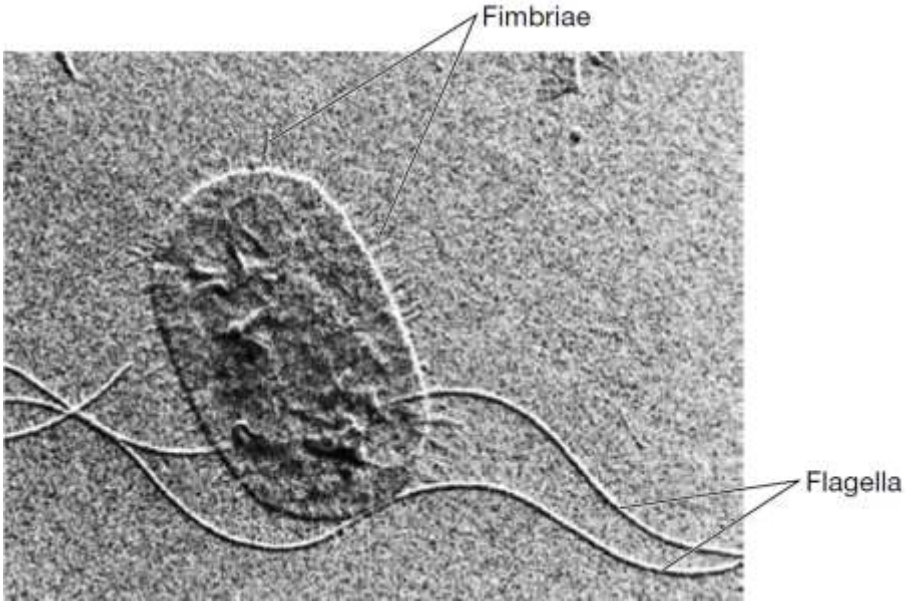
SIGNIFICANCE

- Primarily function is **motility** (chemotaxis, aerotaxis, phototaxis etc).
- **Positive taxis** is movement toward a favorable environment whereas **negative taxis** is movement away from a repellent.
- Flagella can help in **identifying** certain types of bacteria. For example, Proteus species show 'swarming' type of growth on solid media.
- Flagellar **antigens** are used to **distinguish** different species and strains of bacteria
- Variations in the flagellar H antigen are used in **serotyping**.

Important notes: 1. Focus on the structure of the flagella and the classification of bacteria based on the arrangement of flagella.

PILI AND FIMBRIAE

- Fimbriae are short, hair-like structures made up of protein **pilin** and are present in many Gram-negative bacteria.
- Even though pili arise from **plasma membrane** they are not considered part of the plasma membrane.
- They are anchored in the membrane and protrude through the cell wall to the outside of the cell.
- Fimbriae are **shorter** and **straighter** than flagella and are more numerous.
- They are 0.5μm long and 10 nm thick.
- Since they are made up of protein, they are **antigenic**.
- Bacteria from different genera may possess common fimbrial antigens.
- Fimbriae are usually seen in young cultures and lost on subcultures on solid media.
- Sometimes (fimbriae and pili) can be used interchangeably, while the term pili is usually used to denote **sex pili**.
- Sex pili acts to join bacterial cells for transfer of DNA from one cell to another by a process called **conjugation.**



Flagella and Fimbriae. The long flagella and the numerous shorter fimbriae are very evident in this electron micrograph of the bacterium *Proteus vulgaris*

SIGNIFICANCE

- They act as **adhesins** and allow bacteria to colonize cells.
- For example, *Neisseria gonorrhoea* uses its fimbriae to attach to the lining of the genital tract and initiate an infection.
- Fimbriae can also detect chemical **signals** and are important in bacterial cell **communication** and biofilm formation.
- Fimbriae also act as **receptors** for **bacteriophages.**
- Fimbriae of *Streptococcus pyogenes* are coated with **M protein**, which is a virulence factor and adhere to host cells and resist **phagocytosis** (cell eating).
- Fimbriated bacteria form surface pellicle of liquid media.
- Some fimbriae can **agglutinate** RBC of guinea pigs, horses, pigs and fowls. This **hemagglutination** may or may not be inhibited by mannose.

BACTERIAL LOCOMOTION

- While some microbial cells are stationary, some have the ability and the means of getting around. This ability of the bacteria is called the locomotion or motility
- The locomotion in bacteria is of three types; Flagellar, Spirochetal and the gliding movement
- The flagellar movement in bacteria with flagella is powered by the proton energy which results from the movement of ions between the M and the S rings
- This proton energy is then transformed into work which results in the clockwise and anticlockwise movements of the flagella
- These movements then help the bacteria to move from one location to another
- Bacteria like *Pseudomonas, Vibrio, Salmonella, Spirillum* etc. move around with the help of flagellar motility
- The spirochetes like *Treponema, Borrelia, Leptospira* etc. show spirochetal movement.
- These organisms have flagella-like axial filament that is buried in space between inner and outer membranes of the bacterial cell wall
- These filaments are composed of 2 or more fibrils which are embedded in the inner membrane and they act just like the basal body and the motor of the flagella

- The spirochetes can perform flexing, swimming, creeping or spinning type of movements
- Bacteria belonging to the genera *Myxococcus, Cystobacter, Chondromyces* etc. show gliding movements
- While the exact mechanism is still being understood, it is believed that the gliding movements are enabled by fimbriae-like appendages present at the poles in these cells.

CHEMOTAXIS (POSITIVE AND NEGATIVE)

- The phenomenon whereby bacteria and other cells or organisms direct their movements according to certain chemicals in their environment is called chemotaxis.
- Chemotaxis is important particularly in bacteria to find food and to move away from potentially harmful chemicals or poisons (e.g. Phenol)
- If the movement bacteria is toward a higher concentration of the chemical in question then it is called as positive chemotaxis i.e. movement of bacteria towards food. On the other hand, negative chemotaxis occurs if the movement is in the opposite direction or away from potential dangers
- The molecules which trigger chemotaxis are called chemoattractants (positive chemotaxis e.g. glucose) or chemorepellents (negative chemotaxis e.g. Phenol)
- When presented with these chemoattractants or chemorepellents the bacteria with the help of various modes of locomotion will act according to the signal.

INCLUSION BODIES

- Certain **granular structures** lie free within the cytoplasm are present known as inclusion bodies.
- They are granules of organic or inorganic material that often are clearly visible in a light microscope
- These act as **food reserves**, and may contain organic compounds such as starch, glycogen or lipid.
- In addition, sulphur and polyphosphate can be stored as inclusion bodies, the latter being known as **volutin or metachromatic granules**.
- Two types of inclusion body are **Magnetosomes** and **gas vacuoles**

- **Magnetosomes** contain a form of **iron oxide** and help some types of bacteria to orientate themselves downwards into favourable conditions

- **Gas vacuoles** maintain **bouyancy** of the cell in blue greens and some **halobacteria.**